Grade 6 Accelerated
Unit 6
Lesson 1 Practice

Name_______Answer Key____
Date______________________
Period______________________

Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

	Exponential Form	**Expanded Form**	**Value**
1.	$(3.2)^3$	$(3.2)(3.2)(3.2)$	32.768
2.	9^3	$(9)(9)(9)$	729
3.	$\left(\frac{3}{5}\right)^4$	$\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)$	$\frac{81}{625}$
4.	$\left(\frac{5}{2}\right)^5$	$\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}$	$\frac{3125}{32}$

5. What is the value of the expression $(2.8)^2$?

$(2.8)(2.8) = 7.84$

6. Which expression has the larger value, 3^6 or 6^3? Show your work or explain your reasoning.

$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 729$

$6 \cdot 6 \cdot 6 = 216$

$3^6 > 6^3$ since $729 > 216$.

7. Select all expressions that are equivalent to 5^4.

- ☐ $5 + 5 + 5 + 5$
- ☐ 20
- ☑ $5 \cdot 5 \cdot 5 \cdot 5$
- ☑ 625
- ☐ $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$

8. Select all expressions that are equivalent to $\left(\frac{1}{7}\right)^3$.

- ☐ $\frac{3}{7}$
- ☐ 512
- ☐ $\frac{3}{21}$
- ☑ $\frac{1}{7} \cdot \frac{1}{7} \cdot \frac{1}{7}$
- ☑ $\frac{1}{343}$

9. What is the value of the expression $(0.08)^3$?

$(0.08)(0.08)(0.08) = 0.000512$

10. What is the value of the expression $(4.9)^2$?

$(4.9)(4.9) = 24.01$